

AI & ML Task-2 Project Report

Feature Engineering, Model Optimization & Performance Comparison using California Housing Dataset

Objective:

The objective of this project is to apply feature engineering, preprocessing, machine learning model training, optimization, and performance comparison using the California Housing Dataset. Multiple regression models were trained and evaluated to identify the best-performing model for house price prediction.

Methodology

1. Imported required libraries such as pandas, NumPy, matplotlib, and scikit-learn.
2. Loaded the California Housing Dataset using `fetch_california_housing()`.
3. Converted the dataset into a pandas DataFrame.
4. Separated features and target variables.
5. Applied StandardScaler for feature scaling.
6. Split the dataset into training and testing sets.
7. Trained multiple regression models:
 - Linear Regression
 - Ridge Regression
 - Decision Tree Regressor
8. Evaluated models using RMSE and R^2 Score.
9. Compared all models and selected the best-performing model.
10. Visualized Actual vs Predicted House Prices using matplotlib.

Results & Model Comparison

Model	RMSE	R^2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree Regressor	0.724234	0.599732

The Decision Tree Regressor achieved the lowest RMSE and highest R^2 Score, indicating better prediction performance compared to Linear Regression and Ridge Regression models.

Conclusion

This project successfully demonstrated feature engineering, preprocessing, model optimization, and machine learning model comparison using the California Housing Dataset. Feature scaling improved model stability and ensured better training performance. Among all models, the Decision Tree Regressor performed best based on evaluation metrics. The project highlights the importance of data preprocessing and model evaluation in building effective machine learning solutions.

Project Completed Successfully